

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 312 — Past Questions & Materials
by Renance

Course: PEE 312
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 300

This study pack was compiled and restructured by Renance for PEE 312 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 5 — PEE 312 BASIC PETROLEUM ECONOMICS 2024I2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
2024/2025 First Semester Examination
PEE 312: Basic Petroleum Economics

Answer Question One and Any Other Three Questions

Time Allowed: 3 Hours

Please write your matriculation Number boldly on your answer Booklet. Date: 15/05/2025

QUESTION 1 (19 marks)

1(a)i State the categories of probability distribution and mention three examples of each categories.

1a (ii) Define risk management and state three elements of risk management.

(b) A Company is planning to drill 10 exploratory wells with an estimated chance of 20% success. Using Binomial distribution:

(i) What is the probability that the drilling operation will produce exactly six wells?

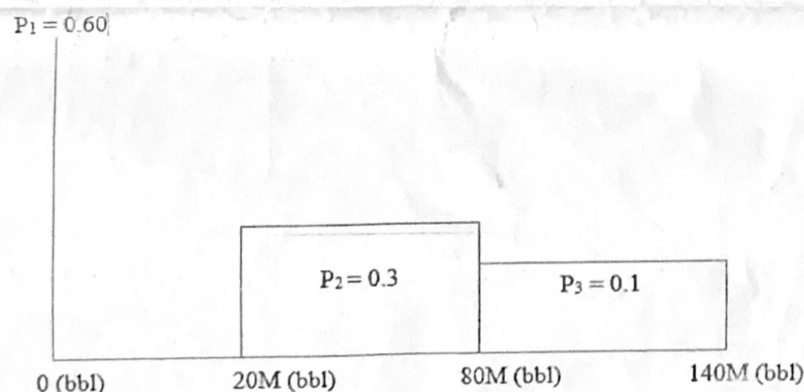
(ii) What is the probability of drilling at least four successful wells?

(iii) What is the probability of drilling at most three successful wells?

Using normal distribution to determine the probability of drilling:

(iv) More than four successful wells.

(c) Determine the mean, variance and standard deviation of the proposed exploratory well shown below.



QUESTION 2 (17 marks)

2(a) What do you understand by the following terms?

- i. Internal Rate of Return (IRR)
- ii. Net Present Value (NPV)
- iii. Pay Back Period (PBP)

iv. Profitability Index (PI)

2(b) An investment produced a series of cash inflows with an initial investment of \$15,000 at 10% interest rate, calculate the:

Cash Flows (\$)	5,000	6,000	3,000	2,500	4,500
Time (years)	1	2	3	4	5

(i) Present Value (ii) Net Present Value (ii) Productivity Index (iv) Discounted Cash flow Rate of Return (iv) Payback Period.

QUESTION 3 (17 marks)

3a (i) Define Petroleum Economics

3a (ii) Define market crude and give four examples.

3b (i) State five importance of petroleum economics.

3b (ii) State five factors which determine the price of crude oil and briefly explain each of the factors.

3c Mention three tools use for evaluating project economic investment and state two importance of each investment tools.

QUESTION 4 (17 marks)

4(a) At what time will \$2,000 amount to \$3,500 at 5%, if the interest rate is compounded quarterly?

(b) Explain the following terms:

(i) Simple Interest (ii) Compound Interest (iii) Present Value (iv) Future Value (v) Annuity.

(c) Calculate the future value at 7% over the period of the life span of the investment.

Cash Flows (\$)	5,000	6,000	3,000	2,500	4,500
Time (years)	1	2	3	4	5

QUESTION 5 (17 marks)

5(a) In January 2018, ABC Ltd. daily expenses were as follow:

Utility cost \$400, production cost \$500, maintenance cost \$200, overhead cost \$250 evaluation cost \$100 and salary payment \$1500.

What is the operating cost of ABC Ltd. at the end of the year?

5b (i) State five examples of capital cost and operating cost.

5b (ii) State and explain the two types of capital expenditure.

5c If an oil producing company spends \$800,000 on fixed asset used in the production of crude oil, \$400,000 in running their daily expenses and the reserve from the oil production is 14,000 stb. What is the unit technical cost?

$$A = P \left(1 + \frac{i}{n} \right)^{nt}$$

$$3500 = 2000 \left(1 + \frac{0.05}{4} \right)^{4t}$$

$$3500 = 2000 (1.0125)^{4t}$$

$$1.75 = 1.0125^{4t}$$

$$\log 1.75 = \log 1.0125^{4t}$$

$$FV = PV \left[\left(1 + \frac{i}{n} \right)^{nt} \right]$$

$$3500 = 2000 \left(1 + \frac{0.05}{4} \right)^{4t}$$

$$1.75 = 1.0125^{4t}$$

$$\log 1.75 = \log 1.0125^{4t}$$

$$A = P \left[1 + \frac{i}{n} \right]^{nt}$$

Paper 2 of 5 — PEE 312 Basic petroleum economics 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
2024/2025 First Semester Examination
PEE 312: Basic Petroleum Economics

A

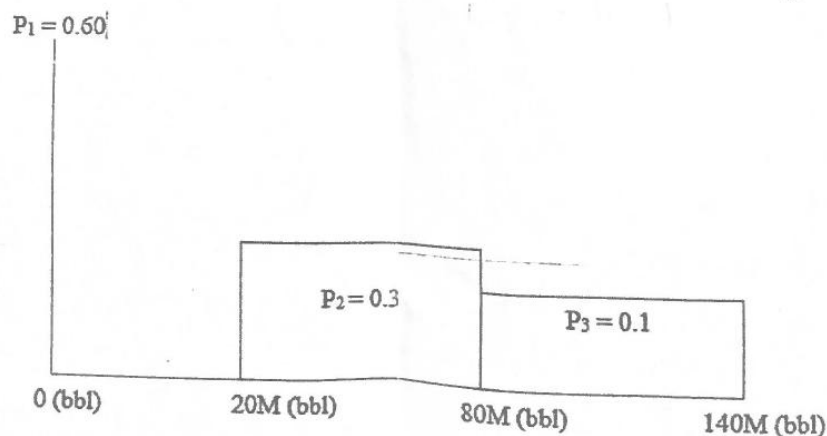
Answer Question One and Any Other Three Questions

Time Allowed: 3 Hours

Please write your matriculation Number boldly on your answer Booklet. Date: 15/05/2025

QUESTION 1 (19 marks)

- 1(a)i State the categories of probability distribution and mention three examples of each categories. ✓
- 1a (ii) Define risk management and state three elements of risk management. ✓
- (b) A Company is planning to drill 10 exploratory wells with an estimated chance of 20% success. Using Binomial distribution:
- (i) What is the probability that the drilling operation will produce exactly six wells? ✓
- (ii) What is the probability of drilling at least four successful wells? ✓
- (iii) What is the probability of drilling at most three successful wells? ✓
- Using normal distribution to determine the probability of drilling: ✓
- (iv) More than four successful wells.
- (c) Determine the mean, variance and standard deviation of the proposed exploratory well shown below. ✓



QUESTION 2 (17 marks)

- 2(a) What do you understand by the following terms?
- Internal Rate of Return (IRR) ✓
 - Net Present Value (NPV) ✓
 - Pay Back Period (PBP) ✓

iv. Profitability Index (PI) ✓

2(b) An investment produced a series of cash inflows with an initial investment of \$15,000 at 10% interest rate, calculate the:

Cash Flows (\$)	5,000	6,000	3,000	2,500	4,500
Time (years)	1	2	3	4	5

(i) Present Value (ii) Net Present Value (ii) Productivity Index (iv) Discounted Cash flow Rate of Return (iv) Payback Period. ✓

QUESTION 3 (17 marks) ✓

3a (i) Define Petroleum Economics

3a (ii) Define market crude and give four examples.

3b (i) State five importance of petroleum economics.

3b (ii) State five factors which determine the price of crude oil and briefly explain each of the factors.

3c Mention three tools use for evaluating project economic investment and state two importance of each investment tools.

QUESTION 4 (17 marks) ✓

✓ 4(a) At what time will \$2,000 amount to \$3,500 at 5%, if the interest rate is compounded quarterly?

(b) Explain the following terms: ✓

(i) Simple Interest (ii) Compound Interest (iii) Present Value (iv) Future Value (v) Annuity. ✓

✓ 4(c) Calculate the future value at 7% over the period of the life span of the investment.

Cash Flows (\$)	5,000	6,000	3,000	2,500	4,500
Time (years)	1	2	3	4	5

QUESTION 5 (17 marks) ✓

✓ 5(a) In January 2018, ABC Ltd. daily expenses were as follow:

Utility cost \$400, production cost \$500, maintenance cost \$200, overhead cost \$250 evaluation cost \$100 and salary payment \$1500.

What is the operating cost of ABC Ltd. at the end of the year?

✓ 5b (i) State five examples of capital cost and operating cost.

✓ 5b (ii) State and explain the two types of capital expenditure.

✓ 5c If an oil producing company spends \$800,000 on fixed asset used in the production of crude oil, \$400,000 in running their daily expenses and the reserve from the oil production is 14,000 stb. What is the unit technical cost?

① Determination of the actual cost of production and services

②

Paper 3 of 5 — PEE 312 Basic Petroleum Economics

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING

2022/2023 First Semester Examination
PEE 312: Basic Petroleum Economics

Answer Questions 1, 2 and 3 any other one from Question 4 and 5. Time Allowed: 3 Hrs.
Please write your Matriculation Number boldly on your answer Booklet. Date: 12/05/2023

Questions

1a (i). List five costing tools necessary for the preparation of accurate capital cost estimate?

(ii). What is the principal objective in economic evaluation and when is an oil or gas project considered economic?

1b. With an initial investment 180 million dollars, the table below shows the Cash flow values obtained by FUPRE Energy Solutions Ltd. in their E & P operations in 5 years.

Year	Revenue \$Millions	OPEX \$Millions	Intangible Drilling Cost \$Millions	Royalty \$Millions	Total Tax Paid \$Millions
1	780	- 120	+ 90	+ 90	+ 335
2	780	- 140	+ 130	+ 90	+ 315.1
3	780	- 135	+ 90	+ 90	+ 332.2
4	780	- 120	+ 90	+ 90	+ 345.1
5	780	- 120	+ 90	+ 90	+ 353.6

Calculate and tabulate the following:

(i). The yearly revenue of the company if its working interest is 100%

(ii). NPV of the net cash flow at 10%, 20%, 30% and at 40% discount rates. $NPV = \frac{NCF}{(1+r)^n}$

(iii). Graph the NPVs of the project as a function of the discount rate.

(iv). Find the IRR from the graph.

2a. Four wells are to be drilled; **BOLO 1, BOLO 2, BOLO 3 and BOLO 4**. The probability of each being successful is 0.5, independent of the outcomes (permutations) of the other wells. In a tabular form, show all possible outcomes, such as **BOLO 1, dry; BOLO 2, producer; BOLO 3, dry; BOLO 4, dry**. Tabulate the results summaries, indicating the probabilities of the various combinations. 1 4 6 4 1

2b (i). Calculate the mean, variance and standard deviation of a uniform distribution with a spike. The uniform distribution extends from 0 to 1,000,000 bbl of oil. The spike is located at 0 and has a probability of 0.4?

(ii). Define and state four characteristics of marker crudes?

Marker crudes can be defined as the mix of the crude oil from different sources and is used to identify the crude oil in the market.

Estimation and prior interest.

$\sum p_i x_i$

- 3a. What is Risk? State any four (4) of such associated with drilling an oil well?
- b (i). Calculate the present value of receiving \$600 at the beginning of the year, five years from now, assuming an annual compounded interest rate of 5%. $F = P(1+i)^n$
- Using the same terms,
- (ii). Calculate the present value of receiving the same amount of money at the midpoint of the fifth year.
- (iii). Calculate the present value of receiving two payments of \$300, one at the beginning of the fifth year and the other at the end assuming the same compounded interest rate.
- 4a.(i). List four different types of petroleum arrangements? Joint Op
- (ii). What are the factors that determine the price of petroleum?
- 4b Give three methods for estimating Capital costs, indicating what the resulting estimate from each method could be used for?
- 4c. Estimate the Operating Expenses for a typical gas plant producing 200 MMSCF/month of gas given the following:
- Operating labour, 6 men each 8 hour shift at \$2.50/hour. $6 \times 8 \times 3 \times 30 \times 2.50$
 - Cost of Supervision \$86,000 annually.
 - Maintenance labour and materials, 7.5% of capital cost (\$10 X 10⁶ capital cost) per annum.
 - Chemical Consumption is as follow:
 - ⊕ 6lb amine/MMSCF of gas processed at 40 cents/lb.
 - ⊕ Drying desiccant at 0.7lb/MMSCF gas processed at 33 cents/lb.
 - ⊕ Absorption oil, 0.1lb/MMSCF of gas processed at 5 cents/lb.
 - ⊕ Sulphur Plant Catalyst $\frac{3}{4}$ lb/ton sulphur processed (150 tons/day sulphur processed catalyst is 10 cents/lb)
- 5a State the five deterministic estimates that must be used for economic viability of a project?
- 5b. A marginal field operator has a 5 years lease on a pipeline facility at one million naira per annum. However, he would like to pay for the lease in one lump sum. If money is worth 1 $\frac{1}{2}$ % per month to the operator, how much should he offer to pay for the facility?
- 5c. You purchased a drilling equipment at the cost of \$9,700,000.00, and made a down payment of \$210,000.00. Your bank financed the balance payment at a nominal annual interest rate of 16%. You are to pay back the loan in 4 years at equal monthly installments.
- How much are you required to pay per month?
 - Repeat question (i) if the bank's interest rate is 9% instead of 19%?

[illegible]

Paper 4 of 5 — PEE 312

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF TECHNOLOGY
Department of Petroleum Engineering

2017/2018 First Semester Post-HND Examinations

PNG 312: Basic Petroleum Economics

Answer Questions 1 & 2 and any other two.

Time Allowed: 2 Hrs.

QUESTIONS

- 1a (i). If as a Petroleum Engineer, acquisition of knowledge of petroleum economics practices in the petroleum industry is desired! Then, state the importance of petroleum economics?
- (ii). When is an oil or gas project considered economic and what is (are) the determinant factors that affects economic viability of a prospective project?
- b. An oil company has to pay \$10,000 per year starting one year from today, on a loan obtained for five years at an effective interest rate of 8%. Calculate the equivalent present value of these five yearly payments. The payments include both principal and interest. Show the cash flow diagram.
- c. Buhari made a bank deposit of ₦50,000 for one month at an annual percentage rate (APR) of 5.875%. Compounded monthly and daily, what is the value after one year?
- 2a. What is Risk? State any four (4) of such associated with drilling an oil well?
- b (i). Calculate the present value of receiving \$500 at the beginning of the year, five years from now, assuming an annual compounded interest rate of 5%.
- Using the same terms in b(i),
- (ii). Calculate the present value of receiving the same amount of money at the midpoint of the fifth year.

- (iii). Calculate the present value of receiving two payments of \$250, one at the beginning of the fifth year and the other at the end assuming the same compounded interest rate.
- c. An oil producer acquired production equipment by paying \$10,000 as down payment and agreed to pay the balance in four year-end installments of \$10,000. Calculate the present value of the annuity due at an interest rate of 8%.
- 3a (i). Explain the term "Time Value of Money".
 (ii). State and explain three (3) factors that determine the price of petroleum?
- b (i). The term "Spot market" is referred to as what? Who is (are) the major player(s) in this kind of market?
 (ii). In five years, it is estimated that \$8,000 will be required to purchase a piece of equipment. What amount should be deposited in a bank today at an interest rate of 7% to accumulate \$8,000 at the end of five years?
- 4a. List four different types of petroleum arrangements?
- b. You purchased drilling equipment at the cost of ₦7,500,000.00, and made a down payment of ₦900,000.00. Your bank financed the balance at an annual interest rate of 16%. If you are to pay back the loan on 70 equal monthly installments.
- (i). How much are you to re-pay per month to pay off the loan in 70?
 (ii). Repeat question (i) if the bank's interest rate is 6% instead of 16%?
- 5a (i). Name any three (3) economic indicators, state two similarities amongst them.
 b. Suppose that \$2,000 is borrowed for one year at an interest rate of 8%. Calculate the interest at the end of the year if the interest is compounded;
 (i). yearly and;
 (ii). quarterly, and;
 (iii). what will be the interest at the end of three years if the interest is compounded semi-annually?

9. i. Already Solved in 2016/2017 Question 1a:

i. Already Solved in 2016/2017 Question 1a ii.

$$\text{From } F = P(1+i)^n$$

when $n=0$, No interest

$$\text{Hence } F = 10,000(1+0.08)^0 = \$10,000.$$

when $n=1$.

$$F = 10,000(1+0.08)^1 = \$10,800.$$

$$\therefore I = F - P = \$10,800 - \$10,000 = \$800.$$

when $n=2$.

$$F = 10,000(1+0.08)^2 = \$11,664$$

$$I = F - P = \$11,664 - \$10,000 = \$1,664.$$

Already Solved in 2016/2017. Question 2a.

Already Solved in 2016/2017. Question 2b.

$$\text{From } A = P \left(\frac{i}{(1+i)^n - 1} + i \right)$$

$$\therefore P = \frac{A}{\left(\frac{i}{(1+i)^n - 1} + i \right)}$$

$$= \frac{\$10,000}{\left(\frac{0.08}{(1+0.08)^4 - 1} + 0.08 \right)}$$

$$P = \$33121.27$$

$$\begin{aligned} \text{Cost of the equipment} &= \text{down payment} + \text{Present Value} \\ &= \$10,000 + \$33121.27 \\ &= \$43121.27 \end{aligned}$$

when $n=3$.

$$F = 10000(1+0.08)^3 = \$12597.12.$$

$$\therefore I = \$12597.12 - 10000 = \$2597.12.$$

when $n=4$

$$F = 10000(1+0.08)^4 = \$13604.89.$$

$$I = \$13604.89 - \$10000 = \$3604.89.$$

when $n=5$.

$$F = 10000(1+0.08)^5 = \$14693.28.$$

$$I = \$14693.28 - \$10000 = \$4693.28.$$



Time value of Money (TVM) is the idea that the Money that is available at the present time is worth more than the same or near same amount in the future due to its potential earning capacity.

To choose the present Money over the future Money is based on economic forecast of:

Inflation rate

Opportunity to invest the present Money

The Risk.

Already Solved in 2016/2017 Question 3a iii.

Already Solved in 2016/2017 Question 3b i

C- from $F = P(1+i)^n$

when compounded Monthly

$$F = 50,000 \left(1 + \frac{0.05875}{12} \right)^{12}$$

$$= \text{₦}53017.9$$

When compounded daily

$$F = 50,000 \times 30 \left(1 + \frac{0.05875}{365} \right)^{365}$$

$$= \text{₦}1590757.60$$

Already solved in 2016/2017 Question 4a.

Already solved in 2016/2017 Question 4b.

Already solved in 2016/2017 Question 69i.

$$F = P(1+i)^n$$

$$I = F - P$$

∴ yearly,

$$F = \$2000(1+0.08) = \$2160.$$

$$I = \$2160 - \$2000 = \$160.$$

Quarterly.

$$F = \$2000\left(1 + \frac{0.08}{3}\right)^{1 \times 3} = \$2164.3.$$

$$I = \$2164.3 - \$2000 = \$164.3.$$

ii.

$$\text{from } A = P \left(\frac{i}{(1+i)^n - 1} \right)$$

$$\therefore i = \frac{0.07}{365} = 0.000192$$

$$\therefore A = 8000 \left(\frac{0.000192}{(1 + 0.000192)^{365 \times 5} - 1} \right)$$

$$\therefore A = \$3.661$$

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF TECHNOLOGY

Department of Petroleum and Natural Gas Engineering

2016/2017 First Semester Examination

PNG 312: Petroleum Economics

Answer Questions 1 & 6 and any other three. Time Allowed: 3 Hrs.

Date: 31/03/2017

QUESTIONS

- 1a(i). As a Petroleum Engineer, acquisition of knowledge of petroleum economics practices in the petroleum industry is desired; state reasons why!
- (ii). When is an oil or gas project considered economic and, what is (are) the determinant factors that affects economic viability of a prospective project?
- b. With an investment of \$200,000,000 in her Nigeria operations; the table below shows the cash flow values over a period 5 years obtained by an international E & P Company, "FEDI Ltd."

Year	Revenue \$Millions	Operating Expenses \$Millions	Royalty \$Millions	Total Tax Paid \$Millions	Intangible Drilling Cost \$Millions
1	800	140	100	355	110
2	800	160	100	335.1	150
3	800	155	100	352.2	110
4	800	130	100	365.1	110
5	800	130	100	373.6	110

Calculate and tabulate the following:

- (i). The yearly revenue of the company if its working interest is 100%.
- (ii). NPV of the net cash flow at 15%, 25%, 35% and at 45% discount rates.
- (iii). Estimate the IRR graphically.
- 2a. What is Risk? State any four (4) of such associated with drilling an oil well?
- b (i). Calculate the present value of receiving \$500 at the beginning of the year, five years from now, assuming an annual compounded interest rate of 5%.
- Using the same terms,
- (ii). Calculate the present value of receiving the same amount of money at the midpoint of the fifth year.
- (iii). Calculate the present value of receiving two payments of \$250, one at the beginning of the fifth year and the other at the end assuming the same compounded interest rate.
- 3a (i). In project cost estimation, under what conditions can an estimate be considered accurate?

- (ii). State and explain three (3) factors that determine the price of petroleum?
- b. (i). The term "Spot market" is referred to as what? Who are the major player(s) in this kind of market?
- (ii). Calculate the mean, variance and standard deviation of a uniform distribution with a spike. The uniform distribution extends from 0 to 100,000 bbl of oil. The spike is located at 0 and has a probability of 0.3?
- 4a. List four different types of petroleum arrangements and explain any two amongst them.
- b. You purchased drilling equipment at the cost of ₦7,500,000.00 and made a down payment of ₦900,000.00. Your bank financed the balance at an annual interest rate of 16%. If you are to pay back the loan on 70 equal monthly installments.
- (i). How much are you to re-pay per month to pay off the loan in 70?
- (ii). Repeat question (i) if the bank's interest rate is 6% instead of 16%?
- 5a. Distinguish between the following terms:
- (i). Profit and Net Cash Flow. (ii). Taxes and Royalties.
- b. Estimate the Operating Expenses for a typical gas plant producing 100 MMSCFD of gas given the following:
- Operating labour, 5 men each 8 hour shift at \$2.0/hour.
 - Cost of Supervision per annum is \$65,000.
 - ~~Annual maintenance labour and materials is 1% of capital cost (\$10 X 10⁶ capital cost).~~
 - Chemical Consumption is as follows:
 - ⊗ 6lb amine/MMSCF of gas processed at 30 cents/lb.
 - ⊗ Drying desiccant at 0.7lb/MMSCF gas processed at 33 cents/lb.
 - ⊗ Absorption oil, 0.1lb/MMSCF of gas processed at 5 cents/lb.
 - ⊗ Sulphur Plant Catalyst 1lb/ton sulphur processed (150 tons/day sulphur processed catalyst is 8 cents/lb).
- 6a. (i). Name any three (3) economic indicators, state two similarities amongst them?
- (ii). List five costing tools necessary for the preparation of accurate capital cost estimate. Explain any two amongst the listed tools.
- c. (i). Four wells are to be drilled: PNB 1, PNB 2, PNB 3 & PNB 4. The probability of each well being successful is 0.5, independent of the outcomes (permutations) of the other wells. Show all possible outcomes, such as PNB 1, dry; PNB 2, producer; PNB 3, dry; PNB 4, dry. Tabulate the results/summaries, indicating the probabilities of the various combinations.
- (ii). A marginal field operator has a 5 years lease on a pipeline facility at one million naira per annum. However, he would like to pay for the lease in one lump sum. If money is worth 12% per month to the operator, how much should he offer to pay for the facility?

1.10 2B

$$F_{\text{from}} F_n = P(1+i)^n$$

Where F_n = Lump Sum

P = Present value

i = Interest rate

n = Period

$$i. \quad P = \frac{F_n}{(1+i)^n}$$

$$P = \frac{500}{(1+0.05)^5} = \$391.76$$

ii. Midpoint of the fifth year implies:

$$n = 4.5$$

$$\therefore P = \frac{500}{(1+0.05)^{4.5}} = \$401.44$$

iii. At the beginning of the 5th year implies:

$$n = 4 \text{ yrs 1 month}$$

$$= 4.0833 \text{ yrs}$$

$$P = \frac{250}{(1+0.05)^{4.0833}} = \$204.84$$

At the end of 5th year implies, $n = 0$

$$P = \frac{-250}{(1+0.09)^0} = \frac{250}{1} = \underline{\underline{\$250}}$$

NO 3

Ai. The accuracy of an estimate depends on the level of experience of the estimator.

ii. Factors that determine the price of petroleum

(i) Market :

Demand and supply plays an important role in the affecting of the price of petroleum. An increase in demand for petroleum would cause an increase in price of petroleum.

(ii) Reliability (production rate) : The number of barrels produced per period determines the price. The higher the number of barrels produced the lower the price of petroleum.

(iii) Location (Transportation) : If the cost of transporting the produced petroleum to where it is needed is high, the price of petroleum tends to be high and if the cost of transportation is low, the reverse is the case.

iv. Quality (refining cost & yield) : If the cost of refining is high, then the price of petroleum increases.

v. Availability (reserves) : The volume of reserves a country has affects the price of petroleum. The higher the reserves, the lower the price of petroleum.

petroleum, the lower the volume of reserves, the higher the price.

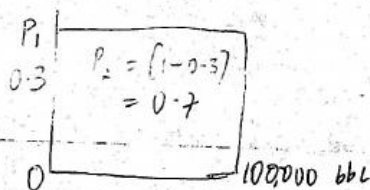
Bc) Spot market refers to one-off sales or spot sales of crude oil in the tanker loads. This is usually crude oil that is surplus to the requirement of the direct purchasers.

Major players in the spot market:

- i. The major international oil companies
- ii. The traders.
- iii. The brokers.
- iv. The independent oil company.

Bm)

SOLUTION



For Part 1 (Spike)

$$P_1 = 0.3, m_1 = 0, \delta_1^2 = 0$$

For Part 2

$$P_2 = 0.7$$

$$n_2 = b_2 + a_2 = \frac{(100,000 + 0)}{2} = 50,000 \text{ bbl}$$

5

$$\sigma_2^2 = \frac{(b_2 - a_2)^2}{12} = \frac{(100,000 - 0)^2}{12} = \frac{(100,000)^2}{12}$$

$$= 8.3 \times 10^7$$

$$M_{\text{total}} = \sum p_i m_i$$

$$= p_1 m_1 + p_2 m_2$$

$$= (0.3 \times 0) + (0.7 \times 50,000)$$

$$M_{\text{total}} = 0 + 35,000$$

$$M_{\text{total}} = 35,000$$

$$\sigma_{\text{total}}^2 = \sum p_i \{ \sigma_i^2 + (m_i - M_{\text{total}})^2 \}$$

$$= p_1 \{ \sigma_1^2 + (m_1 - M_{\text{total}})^2 \} + p_2 \{ \sigma_2^2 + (m_2 - M_{\text{total}})^2 \}$$

$$= 0.3 \{ 0 + (0 - 35,000)^2 \} + 0.7 \{ 8.3 \times 10^7 + (50,000 - 35,000)^2 \}$$

$$\sigma_1^2 = 1,108,333,333 \text{ bbl}$$

$$\sigma = \sqrt{\sigma_1^2}$$

$$= \sqrt{1,108,333,333}$$

$$= 33,291.641 \text{ bbl}$$

NOTE

Types of Petroleum Arrangement

- i. MOU guarantee Margin (Memorandum of Understanding)
- ii. SC (Service Contract)
- iii. JOA (Joint Operating agreement)
- iv. PSC (Production Sharing Contract)

i. MOU [Memorandum of Understanding] =

The M.O.U governs the physical relation between the Nigeria government and J-V partners. The M.O.U guarantees certain level of profit margin to the oil companies irrespective of the fluctuating services.

ii. P.S.C [Production Sharing Contract] =

This arrangement governs the understanding between oil companies and NNPC in the New inland deep and ultra deep water acreages under the PSC =

- i. The company bears all the cost of exploration and production.
- ii. Cost is not reimbursable if no find is made in the acreages.
- iii. Cost is recoverable with crude oil in the event of commercial find.

46)

SOLUTION

Cost of drilling equipment = ₦ 7,500,000.00

Down Payment = ₦ 100,000.00

Amount remaining = ₦ (7,500,000 - 100,000)
= ₦ 6,600,000.00

Hence, $P = ₦ 6,600,000.00$

$$i = \frac{r}{m}$$

where r = Stated annual interest rate

m = How many times a year the investment is compounded from the question, $m = 12$ (monthly)

$$i = \frac{0.16}{12}$$

$$i = 0.0133$$

$$r = 70$$

$$n = m \times r = 12 \times 70 = 70$$

$$A = ₦ 6,600,000 \left(\frac{1 + 0.0133}{(1 + 0.0133)^{70} - 1} \right)$$

$$A = ₦ 145,472.18$$

① For interest rate of 6%

$$i = \frac{0.06}{12} = 0.005$$

$$r = 70$$

$$A = 186,600,000 \left(\frac{0.005}{(1+0.005)^n} \right) \quad | \quad 0.005$$

$$A = 111,979.39$$

NO 5

Profit	Net Cash Flow (NCF)
- Profit does not reflect cash flow - It's an accounting concept used to assess tax liability - It usually incorporates the depreciation of capital cost e.g. cash received - Depreciated CAPEX - OPEX - Royalties - taxes, etc.	- Net cash flow shows the money received and spent - Note used in assessing tax liability - Does not incorporate the depreciation of cash cost e.g. cash received - CAPEX - OPEX - Royalties - Taxes, etc.

② Taxes: This is an important source of revenue in Nigeria. Petroleum Profit tax is paid to J.A.R.C. under TPA.

while it is 50% in psc.

ROYALTIES : These are amount paid to the owner of a resource as compensation for the exploitation of a non-renewable and irreplaceable resource.

(i) Payment is based on volume of oil produced.

(ii) For onshore royalty, it's 20%.

(iii) ~~For~~ Offshore oil royalties decreases as water depth increases from 18.5% - 0% for water depth > 1000m.

B. i. Cost of labour (operating labour) = $5 \times 6 \times 4 \times 2.0$
 $= 240 \text{ /day}$

Nb: Since there are 24 hrs a day and 6 shift is done, hence $24 \div 6 = 4$.

ii. Cost of supervision per day = $\frac{\$65,000}{365} = \178.0
 per day

iii. Maintenance Labour per day = $\frac{5}{100} \times \frac{(10 \times 10^6) \times 1}{365}$
 $= \$1369.86 \text{ per day}$

iv. Chemical consumption per day:

(a) $6 \times 100 \times \frac{30}{100} = \180.0

(b) $0.7 \times 100 \times \frac{30}{100} = \21.0

$$④ \quad \frac{3}{4} \times 150 \times \frac{\%}{100} = 9.0$$

\$ 212.5 per day

$$\text{Total operating cost} = \$ (240 + 179.08 + 1369.85 + 212.5) \text{ per day}$$

$$= \$ 2000.44 \text{ per day}$$

$$\text{Per year} = 2000.44 \times 365$$

$$= \$ 730,160.6 \text{ per year}$$

N10.6

Ai. Three Economic Indication

- ① NPV
- ② PB — Pay Back
- ③ PIR — Profit Investment ratio.

• Similarities between PB and PIR.

- Does not depend on when project starts.
- Does not indicate size of investment.

- Aii.
- Fundamental data
 - Check list
 - Estimating form.
 - Current pricing list.
 - Equipment cost charts & table

④ Estimating form: To standardize estimating procedures and save time, the following are necessary:

- i. Labour rate ii. Electrical and Instrument run rate
- iii. Standard quotation form
- iv. Man hour evaluation.

Current Pricing List: This pertains to individual item cost such as (i) Pipes (ii) Lugs (iii) fittings (iv) pumps (v) valves (vi) etc.

GB.

	WELL 1	WELL 2	WELL 3	WELL 4
All dry	D	D	D	D
1 Producer & 3 dry	P	D	D	D
	D	P	D	D
	D	D	P	D
	D	D	D	P
3 Dry	P	D	D	P
	P	P	D	D
3 Producer	D	P	P	D
	D	D	P	P
	P	D	D	D
	D	P	P	
3 Producer & 1 dry	P	D	P	P
	P	P	D	P
	P	P	P	D
	D	P	P	P
All producer	P	P	P	P

GBii.

SOLUTIONTotal number of outcome = $2^4 = 16$ Expectations (LP)* = D^4 4D³P 6D²P² 4DP³ P⁴

$$\text{Probability} = \frac{\text{No of Outcomes}}{\text{Total possible Outcome}}$$

$$\text{For all dry well} = \frac{1}{16} = 0.0625$$

$$\text{For 1 producer, 3 dry} \\ \text{probability} = \frac{4}{16} = 0.25$$

$$\text{For 3 producer 1 dry} = \frac{4}{16} = 0.25$$

$$\text{For 2 producer 2 dry} = \frac{6}{16} = 0.375$$

$$\text{For all producers} = \frac{1}{16} = 0.0625$$

Tabulating, we have:

Wells	Probabilities
All producers dry	0.0625
1 producer, 3 dry	0.25
2 producer, 2 dry	0.375
3 producer, 1 dry	0.25
All producer	0.0625

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY

2014/2015 First Semester Examination
PNG 312: Petroleum Economics

Answer Question 1 and any other 4 Questions.

Time Allowed: 3 Hours

Please write your Matriculation Number boldly on your answer Booklet.

Questions

- 1a(i). List five costing tools necessary for the preparation of accurate capital cost estimate?
(ii). What is the principal objective in economic evaluation and when is an oil or gas project considered economic?
- 1b. The table below shows the Cash flow values obtained by FearGod Oil and Gas Ltd. in their E & P operations in 5 years.

Year	Investment \$Millions	Revenue \$Millions	Operating Expenses \$Millions	Intangible Drilling Cost \$Millions	Royalty \$Millions	Total Tax Paid \$Millions
0	200					
1		800	140	110	100	355
2		800	160	150	100	335.1
3		800	155		100	352.2
4		800	130	110	100	365.1
5		800	130		100	373.6

Calculate and tabulate the following:

- (i). The yearly revenue of the company if its working interest is 100%.
(ii). NPV of the net cash flow at 15%, 25%, 35% and at 45% discount rates.
(iii). Graph the NPVs of the project as a function of the discount rate.
(iv). Find the IRR from the graph.

- 2a(i). List four different types of petroleum arrangements?
(ii). What are the factors that determine the price of petroleum?
- 2b. Define and state four characteristics of marker crudes?
- 2c. Calculate the mean, variance and standard deviation of a uniform distribution with a spike. The uniform distribution extends from 0 to 1,000,000 bbl of oil. The spike is located at 0 and has a probability of 0.4?
- 3a. Under what conditions does the effective annual rate of interest (EAR) differ from the annual percentage rate (APR)?

$$FV = PV(1+i)^t$$

1,600

John Performance

- 3b. You purchased a drilling equipment at the cost of ₦3,500,000.00, and made a down payment of ₦500,000.00. Your bank financed the balance at an annual interest rate of 16%. You are to pay back the loan on 70 equal monthly installments.
- (i). How much are you to re-pay per month to pay off the loan in 70 months and what will be the total interest payment at the end of the term?
- (ii). Repeat question (i) if the bank's interest rate is 6% instead of 16%?
- 4a. Distinguish between the following terms;
- (i). Risk and Uncertainty
- (ii). Capital and Operating costs.
- 4b. Give three methods for estimating Capital costs, indicating what the resulting estimate from each method could be used for?
- 4c. Estimate the Operating Expenses for a typical gas plant producing 100 MMSCFD of gas given the following:
- Operating labour, 6 men each 8 hour shift at \$2.50/hour.
 - Cost of Supervision \$76,000 annually.
 - Maintenance labour and materials, 5% of capital cost (\$10 X 10⁶ capital cost) per annum.
 - Chemical Consumption is as follows;
 - ⊕ 6lb amine/MMSCF of gas processed at 30 cents/lb.
 - ⊕ Drying desiccant at 0.7lb/MMSCF gas processed at 33 cents/lb.
 - ⊕ Absorption oil, 0.1lb/MMSCF of gas processed at 4 cents/lb.
 - ⊕ Sulphur Plant Catalyst $\frac{2}{3}$ lb/ton sulphur processed (150 tons/day sulphur processed catalyst is 8 cents/lb)
- 5a. State the five deterministic estimates that must be used for economic viability of a project?
- 5b. A marginal field operator has a 3 years lease on a pipeline facility at one million naira per annum. However, he would like to pay for the lease in one lump sum. If money is worth 12.3% per month to the operator, how much should he offer to pay for the facility?
6. Four wells are to be drilled, W, X, Y and Z. The probability of each being successful is 0.5, independent of the outcomes (permutations) of the other wells. In a tabular form, show all possible outcomes; such a W, dry; X, producer; Y, dry; Z, dry. Tabulate the results summaries, indicating the probabilities of the various combinations.

NPV = $\frac{100000(1+i)^t}{(1+i)^t} = 1,125,000$ prob. per $p = \frac{5}{(1+i)^t}$
 $f = p(1+i)^t$
 5900
 $2 \frac{1}{2} = 2.5$
 NPV
 1 1/2 x 1 1/2
 12 per month
 14 months = 14200

2014/2015 SESSION EXAM
 PNG 312: PUBLIC ECONOMY
 PAST QUESTIONS & ANSWERS

18(i) Costing Tools Necessary for the Preparation of Accurate Capital Cost Estimate:

- Fundamental Data
- Check List
- Estimating Forms
- Current Price List
- Installation Cost
- Labour Rate Cost
- Equipment cost charts & Tables
- Cost & Productivity indices
- Technical Preference

Use any (5) above:

- (ii) The principal objective of economic evaluation is to determine the profitability of the project.
- Also an oil and gas project is considered economical if there is a return on the investment of 30% or more.

1b

Note

$$NCF \text{ (Net Cash Flow)} = \text{Revenue} - \text{Expenditure}$$

- i) Therefore, yearly revenue of the company @ 10% interest is the Revenue of that year minus the total expenditure including Royalty and Tax

$$\text{NPV (Net Present Value)} = \sum_{t=1}^n \frac{NCF_t}{(1+i)^t}$$

where

$$NCF = \frac{\text{Revenue} - \text{Expenditure}}{(1+i)^n}$$

i = discount rate

n = No. of years

Table: Completing the table we have:

Year	Investment \$ millions	Revenue \$ millions	Operating Expenses \$ million	Intangible Drilling Cost \$ million	Royalty \$ million
0	200				
1		800	140	110	100
2		800	165	150	160
3		800	155	—	160
4		800	130	110	100
5		800	130	—	100

Total Tax Paid \$ million	NCF \$ millions	PV @ 1 (\$ millions)			
		15%	25%	35%	45%
	-200	-200	-200	-200	-200
355	95	82.61	76	70.37	65.52
235.1	49.9	37.73	31.94	27.38	23.73
352.2	192.8	126.77	98.71	78.36	63.24
265.1	94.9	54.26	38.87	28.57	21.47
273.6	196.4	97.65	64.36	43.80	36.64
		199.02	109.88	48.48	4.60

At 15%, for Year 0

$$PV_0 = \frac{NCF_0}{(1+i)^0} = \frac{-200}{(1+0.15)^0} = \frac{-200}{1} = -200$$

$$PV_1 = \frac{NCF_1}{(1+i)^1} = \frac{95}{(1+0.15)^1} = \frac{95}{1.15} = 82.61$$

Then Repeat for Year 2, 3, 4, 5 to get their respective PV

Therefore:

$$NPV @ 15\% = \sum_{t=0}^{15} PV = (PV_0 + PV_1 + PV_2 + PV_3 + PV_4 + PV_5)$$

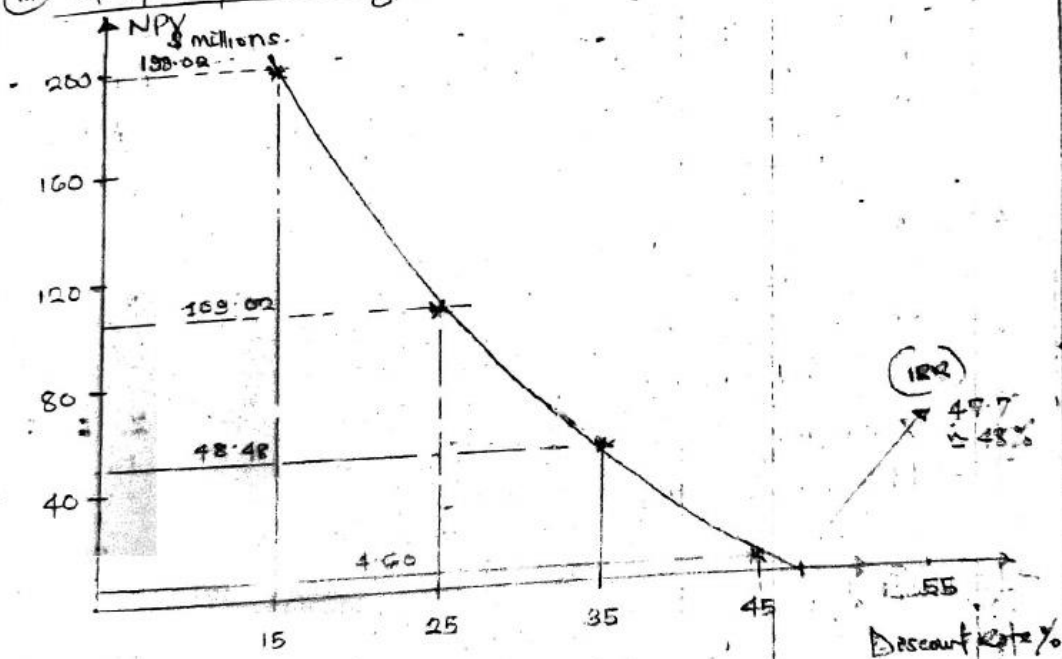
$$NPV @ 15\% \text{ discount rate} = (-200 + 82.61 + 37.73 + 126.77 + 54.26 + 97.65)$$

$$NPV @ 15\% = \$199.02 \text{ millions}$$

Repeat the following steps to find the PV and NPV at 25%, 35% and 45%.

Discount rate %	NPV \$ millions
15	199.02
25	109.88
35	48.48
45	4.60

(iii) Graph of NPV against discount rate.



(iv) From the graph the IRR (Internal Rate of Return) is approximately equal to 48% because it is the discount rate that makes the NPV of a project zero.

2.80 Petroleum Arrangements

- Joint Operating Agreement (JOA)
- Production sharing contract (PSC)
- Service contract
- Memorandum of Understanding (MOU)

(ii): Factors that determine Petroleum Price

- Market (Demand & Supply)
- Reliability (Production Rate)
- Location (Transportation)
- Quality (Refining Cost and Yield)
- Availability (Reserves)

2(b) Marker Crude is an oil from a specified field or region which is traded in spot market and is considered a standard E.g. Bonny Light, Forcados.

Characteristics

- Adequate Reserves
 - Perceived to present "Fair Value"
 - Wide range of buyers and sellers
 - Supply is freely tradable
 - Production is strategically situated
 - Spot prices is widely reported
 - Reasonable immune to manipulation
- Let Any (4) above:

2.c

Take $\beta = 0.4$

Calculating the Mean, Variance & Standard deviation

Part 1 (Spike)

$$M_1 = 0$$

$$\sigma_1^2 = 0$$

Part 2

$$M_2 = \frac{x_2 + b_2}{2} = \frac{0 + 1,000,000}{2} = 500,000$$

$$\sigma_2^2 = \frac{(b_2 - x_2)^2}{12} = \frac{(1,000,000 - 0)^2}{12} = 8.33 \times 10^{10}$$

Therefore

(i) Mean $\bar{x} = \sum P_i M_i$

$$\bar{x} = 0.4 \times 0 + 0.6 \times 500,000$$

$$\bar{x} = 300,000 \text{ bbl}$$

(ii) Variance $\sigma^2 = \sum P_i [\sigma_i^2 + (M_i - \bar{x})^2]$

$$\sigma^2 = 0.4 [0^2 + (0 - 300,000)^2] + 0.6 [8.33 \times 10^{10} + (500,000 - 300,000)^2]$$

$$\sigma^2 = 3.6 \times 10^{10} + 7.398 \times 10^{10}$$

$$\sigma^2 = 10.998 \times 10^{10} \text{ bbl}^2$$

(iii) standard deviation σ

$$\sigma = \sqrt{\sigma^2}$$

$$\sigma = \sqrt{10.998 \times 10^{10}}$$

$$\sigma = 331632.34 \text{ bbl}$$

$$\sigma = 331632 \text{ bbl}$$

3(b). Effective annual rate of interest differs from the nominal percentage rate under the condition that Effective rate of interest takes into account the effect of compounding.

(b) Cost of Drilling Machine = #3,500,000
 Down Payment = #500,000
 Amount Remaining = # (3,500,000 - 500,000)

Annual $P = \#3,000,000$

Interest rate $i = 16\% = 0.16$

Monthly interest = $\frac{0.16}{12} = 0.0133$

$n = 70$ months.

Using $A = P \left(\frac{i}{(1+i)^n - 1} + i \right)$

$A = \#3,000,000 \left(\frac{0.0133}{(1+0.0133)^{70} - 1} + 0.0133 \right)$

$A = \#66024$

Total Interest = Total - (Amount Borrowed)

$= (66024 \times 70) - 3,000,000$

$= \# (4621680 - 3,000,000)$

$= \# 1,621,680$

(ii) Repeating the same for 6% interest rate

$i = 6\% = \frac{0.06}{12} = 0.005$

$n = 70$ months.

Using $A = P \left(\frac{i}{(1+i)^n - 1} + i \right)$

$$A = \text{H } 3,00,000 \left(\frac{100}{100} \right)^{100-1}$$

$$A = \text{H } 50,900$$

Therefore Total interest Payment at the end of the 100th

$$\text{Total Interest} = \text{H}(\Delta \times 100) = \text{H } 3,000,000$$

$$= \text{H } 50,900 \times 100 = \text{H } 3,000,000$$

$$= \text{H } 356,300 - \text{H } 3,000,000$$

$$\text{Total Interest} = \text{H } 563,000$$

4. Distinguish between:

i. Risk & Uncertainty

Basic of Comparison	Risk	Uncertainty
- Meaning	The probability of winning and losing something worthy is known as risk	Uncertainty implies a situation where the future events are not known.
- Ascertainment	It can be measured	Cannot be measured
- Outcome	Chances of outcomes are known	The outcome is unknown
- Control	Controllable	Uncontrollable
- Minimization	Yes	No
- Probabilities	Assigned	Not Assigned

Initial Cost: It is the first cost usually incurred at the beginning of the project. It is a one-time cost and are incurred several times before revenue is obtained. Example of drilling and producing oil well. Operating Cost: It is the regular and necessary costs that go into the operation. This cost only bring benefit in the period that they are incurred. Eg Labour Cost, Well repair Cost etc.

4b. Methods for Estimating Cost

Detailed Cost Estimating Method:

Use: Used by Contractor in bidding for a lump sum Project or job.

Unit Cost Estimating Method:

Use: Used to provide accurate records of past or previous cost experience.

Percent Equipment Cost Estimating Method:

Use: Used as a good preliminary / order of magnitude estimate.

Cost or Labor Factor Cost Estimating Method:

of a project

4c. Gas plant Production rate = 100 MMSCFD

Cost of Labour (Operating Labour)

$$= 100 \times 10 \times 3 \times 2 = \$300 \text{ per day}$$

Cost of Equipment

$$= 100 \times 100 = \$10,000$$

- Maintenance Labour per day

$$= \frac{5}{100} \times 10 \times 10^6 \text{ per year} \times \frac{1}{365}$$

$$= \$ 1,370 \text{ per day}$$

- Chemical Consumption per day

$$\oplus 6 \times 100 \times 0.3 = \$ 180.0$$

$$\oplus 0.7 \times 100 \times 0.33 = \$ 23.1$$

$$\oplus 0.1 \times 100 \times 0.04 = \$ 0.4$$

$$\oplus \frac{3}{4} \times 150 \times 0.08 = \$ 9.0$$

$$\text{Total} = \$ 212.5 \text{ per day}$$

Therefore Total Operating Expenses

$$T_c = \$ (360 + 1208.2 + 1370 + 212.5) \text{ per day}$$

$$= \$ 2150.7 \text{ per day}$$

per year

$$T_c = \$ 2150.7 \times 365$$

$$= \$ 785005.5 / \text{year}$$

5(a) Deterministic Estimates

- Predictive Future Operating Cost
- Failure Analysis
- Funding
- Price Effect / Cost Escalation
- Economic rate / Limit of Production rate

5(b)

Rent per year = #1,000,000

No. of years = 5

Interest Rate = $1\frac{1}{2}\%$ per month = $1\frac{1}{2} \times 12 = 18\%$ per annum $i = 0.18$

Lump Sum = P.V of the end of the fifth year = ?

Year	# Rent	# NCF	# PV
0	0	0	0
1	1,000,000	1,000,000	847,458
2	1,000,000	1,847,458	1,326,816
3	1,000,000	2,326,816	1,416,172
4	1,000,000	2,416,172	1,246,1234
5	1,000,000	2,246,234	981,816

Formula:

$$PV = \frac{NCF}{(1+i)^t}$$

N.B: The Present Value of the previous year plus the Rent of the 4th year gives us the NCF for 4th year

In Conclusion: H. would offer to pay the sum of H981,850.00

G. Given wells W, X, Y, Z. Let Dry = D, Producer = P

W	X	Y	Z	
D	D	D	D	} All Dry
D	D	D	P	
D	D	P	D	
D	P	D	D	
P	D	D	D	} 3 Dry \$ 1 Producer
D	D	P	P	
D	P	D	P	
D	P	P	D	
P	D	P	D	} 2 Dry \$ 2 Producers
P	P	D	D	
P	D	D	P	
P	P	P	D	
P	D	D	P	} 1 Dry \$ 3 Producers
P	P	P	P	
P	D	P	P	
P	P	P	P	
D	P	P	P	} All Producers
P	P	P	P	

Total No. of Outcomes = 16

EVENTS	NO OF OUTCOMES	PROBABILITY
- All Dry	1	$\frac{1}{16} = 0.0625$
- 3 Dry \$ 1 Producer	4	$\frac{4}{16} = 0.25$
- 2 Dry 2 Producers	6	$\frac{6}{16} = 0.375$

1 Dir & 3 Producers	4	$\frac{4}{16} = 0.25$
All Producers	1	$\frac{1}{16} = 0.0625$
TOTAL	16	1.00

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PNG 312: PETROLEUM ECONOMICS

2011/2012: FIRST SEMESTER

TEST

Q1. You have deposited ₦125,000 in the bank paying 13% interest rate compounded annually. At the end of the following eight (8) years, you withdraw the sums indicated in the table. How much will you have in the bank at the end of the eight (8) years? What is the total amount withdrawn from the bank?

End of Year	Interest Earned on balance in the bank during the Year (N)	Balance in the bank at the end of the year or withdrawal	Withdrawal (N)	Balance at the end of the year after withdrawal (N)
0		₦125,000	—	125,000
1	16,250	141,250	38,000	103,250
2	13,422.5	116,672.5	31,000	85,672.5
3	11,137.43	96,809.93	25,000	71,809.93
4	9,385.29	81,145.22	22,000	59,145.22
5	7,688.88	66,834.10	18,000	48,834.10
6	6,348.43	55,182.53	17,000	38,182.53
7	4,963.73	43,146.25	23,000	20,146.25
8	2,619.01	22,765.27	22,750	15.27

Solution.

Since the bank is only compounded annually, the only interest earned is the interest earned in the first year.

Using compound interest formula

$$F_n = P(1+i)^n$$

where

F = future sum of money, P = present sum of money, i = interest rate per period

n = total number of period

At $n=0$ (or before any interest is added to the money)

Balance at the end of the year = \$125,000. It remains the same since no interest has been added, and most importantly, no money has been withdrawn.

At $n=1$ (the end of year 1)

$$F = 125,000(1+0.13)$$

$$= \$141,250 \text{ (Balance before withdrawal)}$$

$$\text{Balance after withdrawal} = \text{Balance b/f withdrawal} - \text{Withdrawal (amount)}$$

$$= 141,250 - 38,000$$

$$= 103,250$$

$$\text{Interest earned on balance} = \text{Balance b/f withdrawal} - \text{Balance after withdrawal the previous year}$$

$$= 141,250 - 125,000$$

$$= 16,250$$

At $n=2$ (The end of year 2)

$$F = F_1(1+i)$$

Balance at year 1

$$= 103,250(1+0.13)$$

$$= 116,675 \text{ (Balance before withdrawal)}$$

$$\text{Balance after withdrawal} = 116,642.5 - 21,000$$

$$= 95,642.5$$

$$\text{Interest earned on balance} = 95,642.5 \times 0.0125$$

$$= 1195.53$$

CONTINUE TO YEAR 8

- ① At the end of the eighth year, I will have \$15.27 in the bank.
- ② Total amount withdrawn from the bank is a summation of all the amount withdrawn.

$$= 19,750$$

Q. An investment of 125,000 for a new road lighting is expected to generate the following cash flow benefits:

End of Year	Cash Flow
1	35,000
2	31,000
3	25,000
4	22,000
5	18,000
6	17,000
7	23,000
8	22,750

What is the PV of this investment at a discount rate of 13%? What would you call this discount rate? Calculate the PIR & DPIR.

Solution

Investment = 125,000.

$$\text{Using } NPV = - \sum_{i=0}^n \frac{NCF_i}{(1+i)^i}$$

$$\text{At the end of year 1} = \frac{35,000}{(1+0.13)} = 30,973.45$$

$$\text{At the end of year 2} = \frac{31,000}{(1+0.13)^2} = 24,277.55$$

$$\text{At the end of year 3} = \frac{25,000}{(1+0.13)^3} = 17,362.5$$

$$\text{At the end of year 4} = \frac{22,000}{(1+0.13)^4} = 13,493.01$$

$$\text{At year 5} = \frac{18,000}{(1+0.13)^5} = 9,769.68$$

$$\text{Year 6} = \frac{17,000}{(1+0.13)^6} = 8,165.41$$

$$\text{Year 7} = \frac{23,000}{(1+0.13)^7} = 9,776.39$$

$$\text{Year 8} = \frac{22,750}{(1+0.13)^8} = 8,557.64$$

$$\begin{aligned} PV &= \sum PV - \text{Investment} \\ &= 125,080.25 - 125,000 \\ &= 80.25 \end{aligned}$$

It will be called Discounted Cashflow Rate of Return (DCFR)
or Internal rate of return (IRR).

$$\begin{aligned} \text{For PIR} \\ \text{N.B.} \quad \frac{\text{Income} - \text{Investment}}{\text{Investment}} \quad \text{OR} \quad \frac{\text{Undiscounted Income} - \text{Investment}}{\text{Investment}} \\ = \frac{196,750 - 128,000}{125,000} = 0.574 \end{aligned}$$

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2011/2012 First Semester Examination
PNG 312: Petroleum Economics

Answer All Questions. Time: 3 Hours
Please write your Matriculation Number Boldly on your Script

- 1) Four wells are to be drilled, A, B, C and D. The probability of each being successful is 0.5, independent of the outcomes (permutations) of the other wells. Prepare a table that shows all possible outcomes, such as A, dry; B, producer; C, dry; D, dry. Summarize the results in a table that shows the probabilities of the various combinations.
- 2a) What are Capital and Operating costs.
- 2b) Give three methods for estimating Capital costs indicating what the resulting estimate from each method could be used for:
- 2c) Estimate the Operating Expenses for a typical gas plant producing ~~100 MM SCF Month~~ ^{100 MM SCF Month} of gas given the following:
- i) Operating labour 5 men each 8 hour shift at \$2.50/hour
 - ii) Cost of supervision \$75,000/year
 - iii) Maintenance labour and materials 5% of Capital cost (\$10 X 10⁶ Capital cost) per year
 - iv) Chemical Consumption is as follows:
 - + 6 lb amine/MMSCF of gas processed at 30 cents/lb
 - + Drying desiccant at 0.7lb/MMSCF gas processed at 32 cents/lb
 - + Absorption oil, 0.1lb/MMSCF of gas processed at 3 cents/lb
 - + Sulphur Plant Catalyst $\frac{1}{4}$ lb/ton Sulphur processed. (150 tons/day Sulphur processed)
Catalyst is 8 cents/lb
- *3a) Calculate the mean, variance and standard deviation of a uniform distribution with a spike. The uniform distribution extends from 0 to 100,000 bbl of oil. The spike is located at 0 and has a probability of 0.3
- 3b) What is the probability that the results will be greater than 40,000 bbl?
- 3c) What is the probability that the results will be greater than 50,000 bbl?
- 4a) Name the two basic types of oil markets and give three characteristics of each of the markets.
- 4b) What are economic indicators? Name four of them.
- *4c) A working interest in a small oil well is expected to generate the following cash flows at the end of each month. What would you be willing to pay for the working interest if money has a time value of 1.3333% per month?

Month	Cash Flow (Mn)
1	100
2	500
8	800
13	700
16	1000

- 5a) A project when completed will yield return payments of N2m each at the ends of the next 10 years. If an apparent return of 25% is required, what is the maximum amount that can be invested in the completed project?
- 5b) What is the corresponding present value of Profit to Investment Ratio at 0% interest rate?

$$\frac{0.98}{1.04} = 0.94$$

1a) What are economic indicators? Name four of them that take time value of money into account.
 1b) A well 5000 ft deep is estimated to produce 400 bopd. Giving the following data, do you recommend a rod pump or an electric submersible pump installation? You have an average investment opportunity rate of 15%.

	Investment	Operating Cost	Life	Salvage Value
Rod Pump	\$42,000	\$8,800/yr	10 yrs	\$5,000
Submersible Pump	\$32,000	\$20,440/yr	10 yrs	\$5,000

5) What is the maximum amount that you can bid for an offshore lease if you have a 20% re-investment opportunity rate using the data below?

Time (Years)	Event
0	Time of bidding
1	Drill exploratory well at a cost of \$2MM
1.5	Drill appraisal well at a cost of \$1.5MM
2	Drill appraisal well at a cost of \$1.5MM
2.5	Begin platform construction at a cost of \$10MM
3	Continue platform construction at a cost of \$10MM
3.5	Install pipeline at a cost of \$2MM
4	Set platform at a cost of \$1MM
4.5	Drill and complete wells at a cost of \$20MM

The Production schedule is as follows:

Time (Years)	Annual Production (bbl)	NCF (\$M)
5	1,300,000	32,500
6	1,300,000	32,500
7	1,100,000	27,500
8	667,000	16,675
9	404,000	10,100
10	245,000	6,125
11	150,000	3,750
12	90,000	2,250
13	50,000	1,250
14	33,000	825
15	20,000	500

At the end of 15 years: Salvage value = Abandonment cost
 What is the DPIR? Assume end of year compounding.

6.) Four wells are to be drilled, A, B, C and D. The probability of each being successful is 0.5, independent of the outcomes (permutations) of the other wells. Prepare a table that shows all possible outcomes, such as A, dry; B, producer; C, dry; D, dry. Summarize the results in a table that shows the probabilities of the various combinations.

$$PIR = \frac{\sum NCF}{\text{Investment}}$$

$$DPIR = \frac{\ln(\text{Sum} - \text{Investment})}{\text{Investment}}$$

Geographic
 Sympathetic
 Deliberate
 Diverse
 Higher selling

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFERUW
 PEG 312: PETROLEUM ECONOMICS

2011/2012 1st SEMESTER EXAMINATION
 SOLUTIONS TO THE QUESTIONS

Q: Four wells named A, B, C, and D.

A dry well; B, producer; C, dry; D, dry.

Probability of each being successful is 0.5, independent of the outcomes (permutation) of other wells.

$$\text{Expectation } (Dp)^4 = D^4, 4D^3p, 6D^2p^2, 4Dp^3, p^4$$

$$\text{Total no of outcomes} = 2^4 = 2^4 = 16$$

Well A	Well B	Well C	Well D	
D	D	D	D	All Dry well
P	D	D	D	3 Dry well, 1 Producer
D	P	D	D	
D	D	P	D	
D	D	D	P	
D	D	P	P	2 Dry, 2 Producer
D	P	D	P	
D	P	P	D	
P	D	D	P	
P	D	P	D	
P	P	D	D	

Q1 well A well B well C well D
P P P P

All Production

Probability of the various combinations

Using Probability = $\frac{\text{no of outcomes}}{\text{Total Possible Outcomes}}$

Total possible outcome = 16

For : All Dry = $\frac{1}{16} = 0.0625$

For : 3 dry, 1 Producer = $\frac{4}{16} = 0.250$

For : 2 dry, 2 Producer = $\frac{6}{16} = 0.375$

For : 1 dry, 3 Producer = $\frac{4}{16} = 0.250$

For : All producer = $\frac{1}{16} = 0.0625$

Tabulating it

Possible Outcomes	Probability
All Dry	0.0625
3 dry, 1 Producer	0.250
2 Dry, 2 Producer	0.375
1 Dry, 3 Producer	0.250
All Producer	0.0625

Q2 (a) Capital Costs: This is a one time cost usually incurred at the beginning of the project.

(b) Operating Costs: These are cost incurred regularly and are necessary to maintain operation.

- i. Let 1 Month = 30.4 days
Recall 1 year = 12 Months
- i. $5 \times 8 \times \$2.5 = \100
But, this \$100 is just for 8 hours and not a day
 $8 \times \$100 = \800
 $\therefore$ for a Month = $30.4 \times \$300 = \$9120/\text{Month}$
- ii. Cost of Supervision = \$75,000/year
But for a Month, we divide by 12.
$$= \frac{\$75,000}{12} = \$6,250/\text{Month}$$
- iii. Maintenance of labour and Material at 5% of Capital Cost = $\$10 \times 10^6/\text{year}$
$$= \frac{5}{100} \times 10 \times 10^6$$

$$= \$5 \times 10^5/\text{year}$$

for a Month, divide by 12:
$$= \frac{\$5 \times 10^5}{12} = \$41,666.67/\text{Month}$$
- iv. Chemical Consumption per Month,
- * $6 \times 100 \times 0.3 = \180
 - * $0.7 \times 10 \times 0.32 = \22.4
 - * $0.1 \times 100 \times 0.003 = \0.3
 - * $3/4 \times 150 \times 30.4 \times 0.08 = \273.6
- Total = $\$476.3/\text{Month}$

Therefore total Operating Expenses

$$T_o = \$ (9120 + 6250 + 41666.67 + 476.3) \text{ per Month}$$

$$= \$ 57512.97 \text{ per Month}$$

per year:

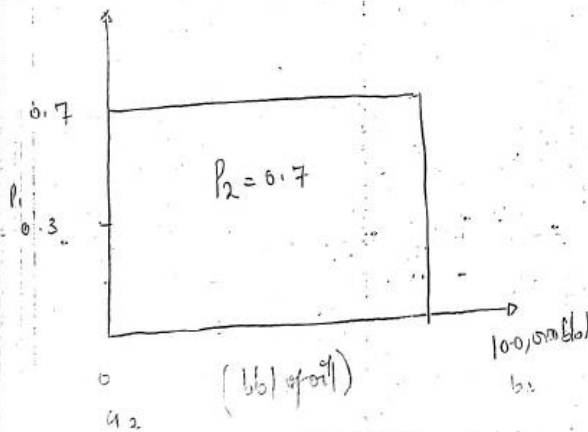
$$T_o = \$ 57512.97 \times 12$$

$$= \$ 690155.64 / \text{year}$$

Q5. Calculate the mean, variance, and standard deviation of a uniform distribution with a spike. The uniform distribution extends from 0 to 100,000 bbl of oil. The spike is located at 0 and has a probability of 0.3.

- (b) What is the probability that the results will be greater than 40,000 bbl?
 (c) What is the probability that the results will be greater than 50,000 bbl?

Solution.



$m = \text{mean}$, $\sigma^2 = \text{variance}$, $\sigma = \text{standard deviation}$

Probability of spike = 0.3

Probability of uniform distribution = $1 - 0.3 = 0.7$

Part 1: Spike (1st part)

$$m_1 = 0$$

$$\sigma_1^2 = 0$$

Uniform distribution (2nd part)

$$\sigma_x = \frac{(100,000 - 0)^2}{12} = 833,333,333.33$$

Finding the mean, variance & standard deviation.

$$\text{Mean} = \sum p_i m_i$$

$$= 0.3(0) + 0.7(50,000)$$

$$= 0 + 35,000$$

$$= 35,000$$

Variance

$$= \sum p_i \{ \sigma_i^2 + (m_i - m)^2 \}$$

$$= 0.3 \{ (0^2 + (0 - 35,000)^2) + (833,333,333.33 + (50,000 - 35,000)^2) \}$$

$$= 1,108,333,333.33$$

$$\sigma = \sqrt{\sigma^2}$$

$$= \sqrt{1,108,333,333.33}$$

$$= 33,291.47$$

$$\text{b) Probability} = \frac{\text{Possible Outcome}}{\text{Total Outcome}}$$

$$= \frac{40,000}{100,000}$$

$$= 0.4$$

The probability that the result will be greater than 40,000 bbl

$$= 0.7 - 0.4$$

$$= 0.3$$

$$\begin{aligned} \textcircled{c} \quad & \frac{50,000}{100,000} \\ & = 0.5 \end{aligned}$$

The prob. that the results will be greater than 50,000 is

$$\begin{aligned} & 0.7 - 0.5 \\ & = 0.2 \end{aligned}$$

- ④ (i) The "Net" or "Cash" Market
(ii) Future Market

④ Economic Indicators: These are economic tools used to determine the viability of a given project.

- (i) Predictive future operating cost
- (ii) future analysis
- (iii) funding
- (iv) price effect (cost escalation)

$$\textcircled{A} \quad i = 1.333\% = 0.01333$$

Month	Cash flow (M)
1	-150

Solution

Using $\left[\frac{NCF}{(1+i)^j} \right]$ Note: "-" implies losses
"+" implies profit

Month	Cash flow	At 1.333% Per month
1	-100	-98.7
2	+500	487
8	+800	719.6
13	-700	-589.3
16	+1000	809.03
Σ (Summation)		1327.63

Profits or Losses

Month 1

$$\frac{-100}{(1+0.01333)^1} = -98.7 \text{ (Loss)}$$

$$\frac{+1000}{(1+0.01333)^{16}} = 809.03$$

Sum of Losses

$$-98.7 - 589.3 = -688$$

Sum of Profits

$$487 + 719.6 + 809.03$$

$$= 2015.63$$

 Σ Profit + Σ of Losses

$$2015.63 - 688$$

$$= 1327.63 //$$

Month 2

$$\frac{+500}{(1+0.01333)^2} = 487 \text{ (Profit)}$$

Month 3

$$\frac{+800}{(1+0.01333)^8} = 719.6 \text{ (Profit)}$$

Month 4

$$\frac{-700}{(1+0.01333)^{13}} = -589.3 \text{ (Loss)}$$

Using

$$NPV = \left[\frac{NCF}{(1+i)^j} \right]$$

$$i = 25\% = 0.25$$

$$j = \text{no. of years (t)}$$

$$NCF = 2 \times 10^6$$

$$\text{For 1st year} = \frac{2,000,000}{(1+0.25)}$$

$$= \frac{2,000,000}{1.25}$$

$$= 1,600,000$$

$$\text{For year 2} = \frac{2,000,000}{(1+0.25)^2}$$

$$= \frac{2,000,000}{1.5625}$$

$$= 1,280,000$$

Continue till

$$\text{Year 10} = \frac{2,000,000}{(1+0.25)^{10}}$$

Table

End of Year	Return/(CF)	Value at 25%
1	2×10^6	1,600,000
2	"	1,280,000
3	"	1,024,000
4	"	819,200
5	"	655,360
6	"	524,288
7	"	419,430.4
8	"	335,544.32
9	"	268,435.5
10	2×10^6	214,748.95
Σ (Summation)		7,141,007.16

The maximum amount that can be invested in the complete project is \$714,007.16.

(b) Corresponding Present value of Profit to Investment Ratio at 0% Interest rate?

NR: 20,000,000

1. $E(X) = \sum x_i p_i = 1 \cdot \frac{1}{16} + 2 \cdot \frac{4}{16} + 3 \cdot \frac{7}{16} + 4 \cdot \frac{4}{16} + 5 \cdot \frac{1}{16} = \frac{1}{16} + \frac{8}{16} + \frac{21}{16} + \frac{16}{16} + \frac{5}{16} = \frac{51}{16} = 3.1875$
 2. $E(X^2) = \sum x_i^2 p_i = 1^2 \cdot \frac{1}{16} + 2^2 \cdot \frac{4}{16} + 3^2 \cdot \frac{7}{16} + 4^2 \cdot \frac{4}{16} + 5^2 \cdot \frac{1}{16} = \frac{1}{16} + \frac{16}{16} + \frac{63}{16} + \frac{64}{16} + \frac{25}{16} = \frac{170}{16} = 10.625$
 3. $\sigma^2 = E(X^2) - [E(X)]^2 = 10.625 - (3.1875)^2 = 10.625 - 10.16015625 = 0.46484375$
 4. $\sigma = \sqrt{0.46484375} \approx 0.6818$

(b) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(c) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(d) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(e) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(f) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(g) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(h) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(i) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(j) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(k) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(l) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(m) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(n) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(o) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(p) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(q) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(r) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(s) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(t) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(u) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(v) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(w) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(x) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(y) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

(z) $\sigma^2 = 0.46484375$
 $\sigma = 0.6818$

108,333,333.57

The expected value is 0

$= 38,291,666.58$

B) Probability = $\frac{\text{possible outcome}}{\text{Total outcome}}$

$= \frac{40,000}{100,000}$

$= 0.4$

The probability of results will be $> 40,000$

$= 0.7 - 0.4$

$= 0.3$

C) Probability = $\frac{\text{possible outcome}}{\text{Total outcome}}$

$= \frac{50,000}{100,000}$

$= 0.5$

The probability of results will be $> 50,000$

$= 0.5$

D) As seen in Part C

4

Expected value of money has a time value

1.3333

Time	Cash flows (thousands)
1	-100
2	+500
3	+800
4	-700
5	+1000

Expected value from Pascal's experiment

$(0.5)^4 = (0.5^4)(40^3P^2)(60^2P^2)(40P^3)(1P^4)$

No. of outcomes $2^4 = 2^4 = 16$

Time	Cash flows (thousands)	Expected value
1	-100	-78.7
2	+500	487
3	+800	719.6
4	-700	-579.3
5	+1000	869.63
Total		1,327.63

Using NCF

$(1+0.1)$

The positive sign implies profits

The negative sign implies losses

② PROFITS

$\frac{500}{(1+0.01333)^2} = 487$

$\frac{800}{(1+0.01333)^3} = 719.6$

$\frac{1000}{(1+0.01333)^4} = 869.63$

$\frac{700}{(1+0.01333)^5} = 579.3$

$\frac{100}{(1+0.01333)^6} = 93.3$

$\frac{100}{(1+0.01333)^7} = 93.3$

$\frac{700}{(1+0.01333)^8} = 579.3$

$\frac{100}{(1+0.01333)^9} = 93.3$

Profit + Losses

$(1,327.63) - (0) = 1,327.63$

15
The general volume completed will yield at
least 30 to 40 million each all the units of the
country. I am not sure of the exact number
of units which is the maximum minimum
available that can be involved in a complete
project.

(5) What is corresponding present value of Profit to Investment Ratio at 0% interest rate?

SOLUTION		
End of Year	Rate (%)	At 25 Years
1	2×10^6	1,600,000
2	2×10^6	1,280,000
3	2×10^6	1,024,000
4	2×10^6	819,200
5	2×10^6	655,360
6	2×10^6	524,288
7	2×10^6	419,430.4
8	2×10^6	335,544.32
9	2×10^6	268,435.5
10	2×10^6	214,748.4
	2×10^6	7,141,006.61

$$= \frac{2,000,000}{(1 + 0.25)^1}$$

③ The maximum amount heard can be in the uncompleted project

D-7-141,006-6

3) Protein measurement - Post - 0.6 ml

$$\text{PIR} = \frac{20,000,000}{7,141,006} = 2.7, 141, 00$$

1.1

② 2014-2-19-2015-1

① It is not possible to determine whether the
statement is true or false because a company may
not have available information to measure its products
and services available at the time of the proposed treatment.
Study may be realized as a result of proposed treatment.
Calculation of net present value at each outcome level
and a value and then mean variance & standard
deviation of 3 periods that present value outcome.

Possible Value of Investment Return (x)	Probability
400,000	0.25
600,000	0.10
800,000	0.30
1,000,000	0.20
1,200,000	0.15

SOLUTIONS	Probability (P _i)	Net Present Value
Invested below weighted Products		
0	0.25	-500,000
400,000	0.10	-100,000
800,000	0.30	100,000
1,200,000	0.20	300,000
1,600,000	0.15	500,000
		NPV = 200,000

$41,470 - 90 = \text{will be added to construction}$
 $= 0 + 589,800$
 $= -589,800$
 $41,400,000 - 400,000 - 500,000$
 $= 40,900,000$

P_i	X_i	$P_i X_i$	X_i^2	$P_i X_i^2$
4500,000	0.25	1,125,000	0.0625	281,250
1100,000	0.10	110,000	0.0100	11,000
100,000	0.10	10,000	0.0100	1,000
300,000	0.20	60,000	0.0400	12,000
300,000	0.20	60,000	0.0400	12,000
500,000	0.15	75,000	0.0225	11,250
ΣP_i		70,000		121,000

$\mu_{\text{neu}} = 10.000 - 207 = 9793$
 $\text{Varianz} = (21.41 - 10)^2 = 129.7$
 $\text{Standardabweichung} = \sqrt{129.7} = 11.39$
 $21.41 - 11.39 = 10.02$
 $10.02 - 11.39 = -1.37$

Labor: $2 \times 10 \times 8 \times 10.50 = 2,520$
 Supervisor/day: $\frac{750,000}{305} = 2,059.8$
 Maintenance: $0.05 \times 100 \times 10^6 = 5,000,000$
 Labor cost/day: $5,000,000 = 13,698.0$
 Churned: $12 \times 200 \times 0.5 = 3,000$
 Conversion: $0.1 \times 500 \times 0.42 = 199$
 Absorption: $0.2 \times 500 \times 0.05 = 8$
 Subsidy: $1.0 \times 200 \times 0.2 = 40$
 Catalyst: $1.0 \times 200 \times 0.2 = 40$
 Total cost per day: $2,520 + 2,059.8 + 13,698.0 + 3,000 + 199 + 8 + 40 = 24,465.4$
 Total cost: $24,465.4 \times 305 = 7,361,841.7$

(3)
 Using distribution with probability of 0.6 and 0.4, find the mean and variance. The mean is calculated as follows:
 $\mu = 0.6(-6.5 \times 10^6) + 0.4(14 \times 10^6)$
 $\mu = -3.9 \times 10^6 + 5.6 \times 10^6 = 1.7 \times 10^6$
 The variance is calculated as follows:
 $\sigma^2 = 0.6(-6.5 \times 10^6)^2 + 0.4(14 \times 10^6)^2 - (\mu)^2$
 $\sigma^2 = 0.6(42.25 \times 10^{12}) + 0.4(196 \times 10^{12}) - (1.7 \times 10^6)^2$
 $\sigma^2 = 25.35 \times 10^{12} + 78.4 \times 10^{12} - 2.89 \times 10^{12} = 100.86 \times 10^{12}$
 Standard deviation: $\sigma = \sqrt{100.86 \times 10^{12}} = 10.04 \times 10^6$

(4)
 Using distribution with probability of 0.6 and 0.4, find the mean and variance. The mean is calculated as follows:
 $\mu = 0.6(-6.5 \times 10^6) + 0.4(14 \times 10^6)$
 $\mu = -3.9 \times 10^6 + 5.6 \times 10^6 = 1.7 \times 10^6$
 The variance is calculated as follows:
 $\sigma^2 = 0.6(-6.5 \times 10^6)^2 + 0.4(14 \times 10^6)^2 - (\mu)^2$
 $\sigma^2 = 0.6(42.25 \times 10^{12}) + 0.4(196 \times 10^{12}) - (1.7 \times 10^6)^2$
 $\sigma^2 = 25.35 \times 10^{12} + 78.4 \times 10^{12} - 2.89 \times 10^{12} = 100.86 \times 10^{12}$
 Standard deviation: $\sigma = \sqrt{100.86 \times 10^{12}} = 10.04 \times 10^6$

Paper 5 of 5 — PEE 312

Level 300 • First Semester • Handout

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PAYBACK (PB) P1-

- 1) Indicates length of investment "exposure" or break-even point of a project.
- 2) Easy to calculate and understand.
- 3) It Ignores

Not Discounted

using discounted cash flow

Cash flow opportunity rate, profit-to-investment rate,
 That rate which helps to choose our choice earlier ahead

moments where you have multiple present value.

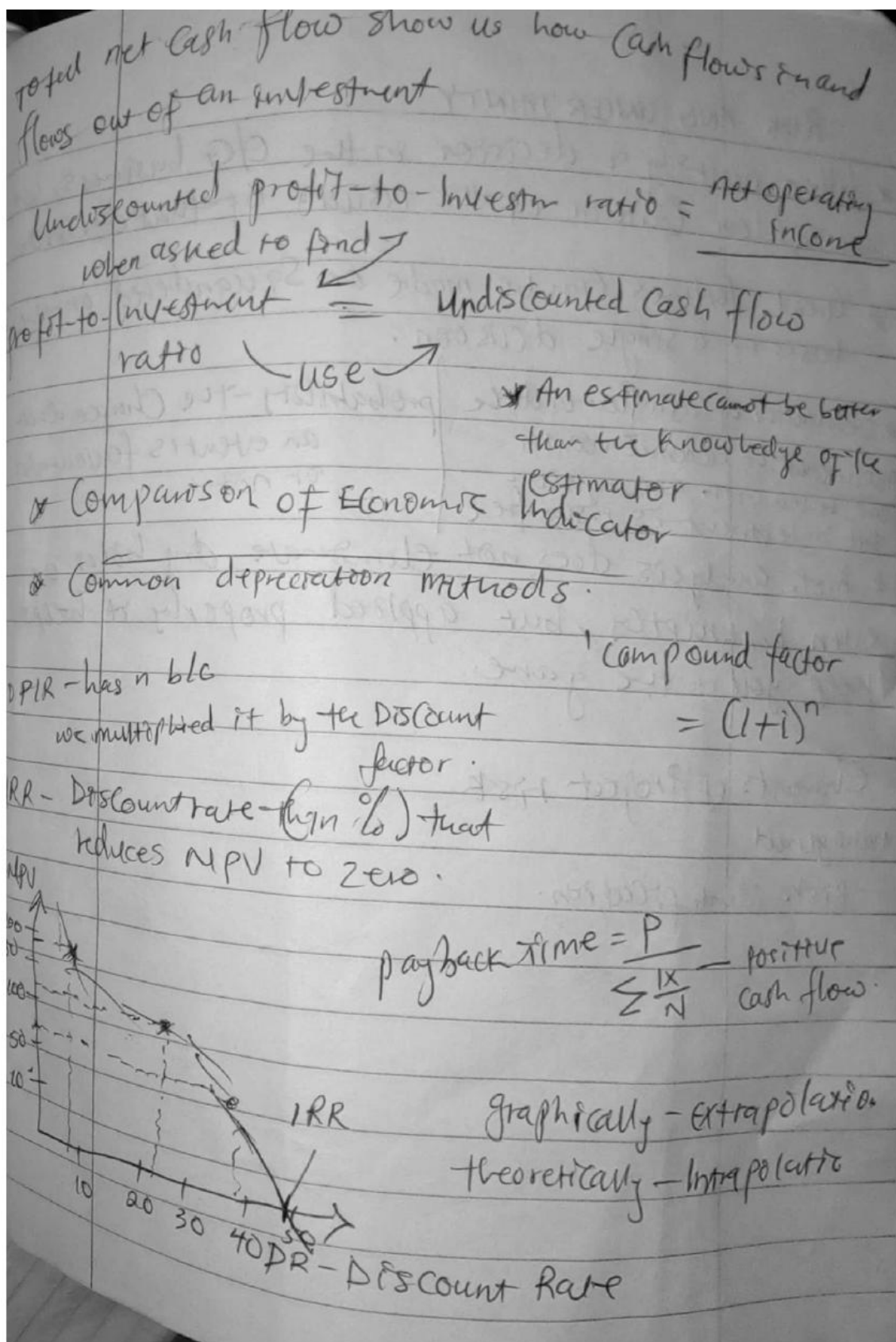
That rate which helps to choose our choice earlier ahead

A project is only economical when we can ∇

Add additional unit of investment or utility.

Example

- * Total undiscounted net operating income = \$585,367.
- * Total undiscounted investment = \$225,000
- * Total undiscounted net cash flow =
- * Total investment discounted at 10% = \$425,000
(because only one investment was made and that was at time 0).
- * Total net cash flow discounted at 10% = \$99,368.
- * Total net operating income discounted at 10% = \$99,368 + \$425,000
= \$524,368.



RISK AND UNCERTAINTY

* When making a decision in the O/G business, we are seldom certain of the results of that decision.

* Great fortunes can be made or squandered on the basis of a single decision.

* Deviation of cash flow and the measurement of economic worth are based on the assumption that investment is risk-free

probability - the chance whether an event is favourable or not.

* Risk analysis does not eliminate dry holes or even bankruptcies, but applied properly it helps keep you in the game.

Elements of Project risk management

— Risk Identification.